平面波的反射与折射

- 平面波射至光学性质不同的2个媒介界面上，分为透射波与反射波。

- 平面波，沿单位矢量 $s^{(i)}$ 方向传播，若知其在空间某点时间行为，波便完全确定。已知 $F(t)$，任一点表为 $F(t-\frac{\vec{r}\cdot\vec{s}}{v})$

若令 $s^{(r)}$ 与 $s^{(t)}$ 表反射、折射波传播方向上单位矢量，令3波函数在界面上矢量相等

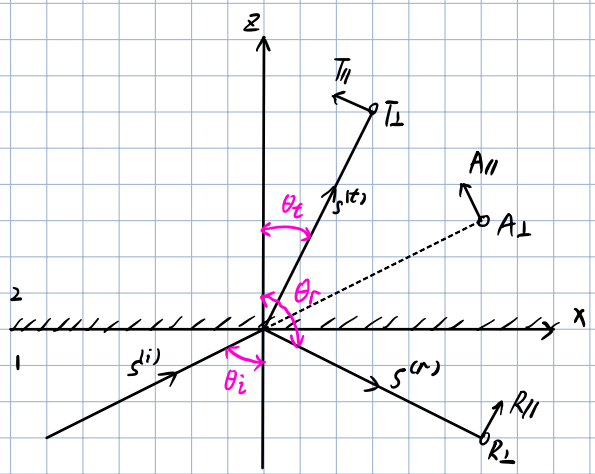
$$t-\frac{r\cdot s^{(i)}}{v_1} = t-\frac{r\cdot s^{(r)}}{v_1} = t-\frac{r\cdot s^{(t)}}{v_2}$$

直角坐标表示，$r\equiv x、y、0$

$$\frac{xs_x^{(i)}+ys_y^{(i)}}{v_1} = \frac{xs_x^{(r)}+ys_y^{(r)}}{v_1} = \frac{xs_x^{(t)}+ys_y^{(t)}}{v_2}$$

for $\forall x$，上式成立，therefore

$$\begin{cases} \frac{s_x^{(i)}}{v_1} = \frac{s_x^{(r)}}{v_1} = \frac{s_x^{(t)}}{v_2} \\ \frac{s_y^{(i)}}{v_1} = \frac{s_y^{(r)}}{v_1} = \frac{s_y^{(t)}}{v_2} \end{cases} \quad (3)$$



如图，以入射面为xz平面，用 θ_i、θ_r、θ_t 表 $s^{(i)}$、$s^{(r)}$、$s^{(t)}$ 与 Oz 所成角度.

有：

$$\begin{cases} s_x^{(i)} = \sin\theta_i & s_y^{(i)} = 0 & s_z^{(i)} = \cos\theta_i \\ s_x^{(r)} = \sin\theta_r & s_y^{(r)} = 0 & s_z^{(r)} = \cos\theta_r \\ s_x^{(t)} = \sin\theta_t & s_y^{(t)} = 0 & s_z^{(t)} = \cos\theta_t \end{cases} \quad (4)$$

代入(3)式，有：

$$\frac{\sin\theta_i}{v_1} = \frac{\sin\theta_r}{v_1} \longrightarrow \theta_r = \pi - \theta_i \quad \text{反射定律}$$

$$= \frac{\sin\theta_t}{v_2} \longrightarrow \frac{\sin\theta_i}{\sin\theta_t} = \frac{v_1}{v_2} = \frac{\sqrt{\varepsilon_2\mu_2}}{\sqrt{\varepsilon_1\mu_1}} = n_{12} \quad \text{折射定律}.$$

全反射？

when $n_2 > n_1$，$n_{12} > 1$

$$\sin\theta_t = \frac{1}{n_{12}}\sin\theta_i < \sin\theta_i$$

only when $\sin\theta_i \leq n_{12}$，θ_t 有实值.

菲涅尔公式.

以上仅考虑相位，现考虑振幅.

假定2介质均匀各向同性、完全透明(电导率为0，磁导率为1)

设A为入射场电矢量振幅，含其位相为波函数常量常数部份。变数部分为:

$$\tau_i = \omega\left(t - \frac{r\cdot s^{(i)}}{v_1}\right) = \omega\left(t - \frac{x\sin\theta + z\cos\theta_i}{v_1}\right)$$

将场矢量分解为//与⊥入射面，入射场电矢量各分量:

$$\begin{cases} E_x^{(i)} = -A_{//}\cos\theta_i e^{-i\tau_i} \\ E_y^{(i)} = A_{\perp} e^{-i\tau_i} \\ E_z^{(i)} = A_{//}\sin\theta_i e^{-i\tau_i} \end{cases}$$

磁矢量

$$H = \sqrt{\varepsilon}\,\vec{s} \times \vec{E}$$

∴

$$\begin{cases} H_x^{(i)} = -A_{\perp}\cos\theta_i\sqrt{\varepsilon_1}\, e^{-i\tau_i} \\ H_y^{(i)} = -A_{//}\sqrt{\varepsilon_1}\, e^{-i\tau_i} \\ H_z^{(i)} = A_{\perp}\sin\theta_i\sqrt{\varepsilon_1}\, e^{-i\tau_i} \end{cases}$$

同理，设T、R为透射、反射波振幅，相应分量:

透射场:

$$\begin{cases} E_x^{(t)} = -T_{//}\cos\theta_t e^{-i\tau_t},\quad E_y^{(t)} = T_{\perp} e^{-i\tau_t},\quad E_z^{(t)} = T_{//}\sin\theta_t e^{-i\tau_t} \\ H_x^{(t)} = -T_{\perp}\cos\theta_t\sqrt{\varepsilon_2}\, e^{-i\tau_t},\quad H_y^{(t)} = -T_{//}\sqrt{\varepsilon_2}\, e^{-i\tau_t},\quad H_z^{(t)} = T_{\perp}\sin\theta_t\sqrt{\varepsilon_2}\, e^{-i\tau_t} \\ \tau_t = \omega\left(t - \frac{r\cdot s^{(t)}}{v_2}\right) = \omega\left(t - \frac{x\sin\theta_t + z\cos\theta_t}{v_2}\right) \end{cases}$$

反射场：

$$\begin{cases} E_x^{(r)} = -R_{//}\cos\theta_r e^{-i\tau_r}, \quad E_y^{(r)} = R_\perp e^{-i\tau_r}, \quad E_z^{(r)} = R_{//}\sin\theta_r e^{-i\tau_r} \\ H_x^{(r)} = -R_\perp\cos\theta_r\sqrt{\varepsilon_2}e^{-i\tau_r} \quad H_y^{(r)} = -R_{//}\sqrt{\varepsilon_2}e^{-i\tau_r}, \quad H_z^{(r)} = R_\perp\sin\theta_r\sqrt{\varepsilon_2}e^{-i\tau_r} \\ \tau_r = \omega\left(t - \frac{r\cdot s^{(r)}}{v_1}\right) = \omega\left(t - \frac{x\sin\theta_r + z\cos\theta_r}{v_1}\right) \end{cases}$$

由边界条件切线分量连续性：

$$\begin{cases} E_x^{(i)} + E_x^{(r)} = E_x^{(t)} \qquad E_y^{(i)} + E_y^{(r)} = E_y^{(t)} \\ H_x^{(i)} + H_x^{(r)} = H_x^{(t)} \qquad H_y^{(i)} + H_y^{(r)} = H_y^{(t)} \end{cases} \tag{18}$$

B、D 法线分量自动满足.

将所有分量代入 (18)，由 $\cos\theta_r = \cos(\pi - \theta_i) = -\cos\theta_i$，有：

$$\begin{cases} \cos\theta_i(A_{//} - R_{//}) = \cos\theta_t T_{//} \\ A_\perp + R_\perp = T_\perp \\ \sqrt{\varepsilon_1}\cos\theta_i(A_\perp - R_\perp) = \sqrt{\varepsilon_2}\cos\theta_t T_\perp \\ \sqrt{\varepsilon_1}(A_{//} + R_{//}) = \sqrt{\varepsilon_2} T_{//} \end{cases}$$

解得：

$$T_{//} = \frac{2n_1\cos\theta_i}{n_2\cos\theta_i + n_1\cos\theta_t} A_{//}$$

$$T_\perp = \frac{2n_1\cos\theta_i}{n_1\cos\theta_i + n_2\cos\theta_t} A_\perp$$

$$R_{//} = \frac{n_2\cos\theta_i - n_1\cos\theta_t}{n_2\cos\theta_i + n_1\cos\theta_t} A_{//}$$

$$R_\perp = \frac{n_1\cos\theta_i - n_2\cos\theta_t}{n_1\cos\theta_t + n_2\cos\theta_t} A_\perp$$

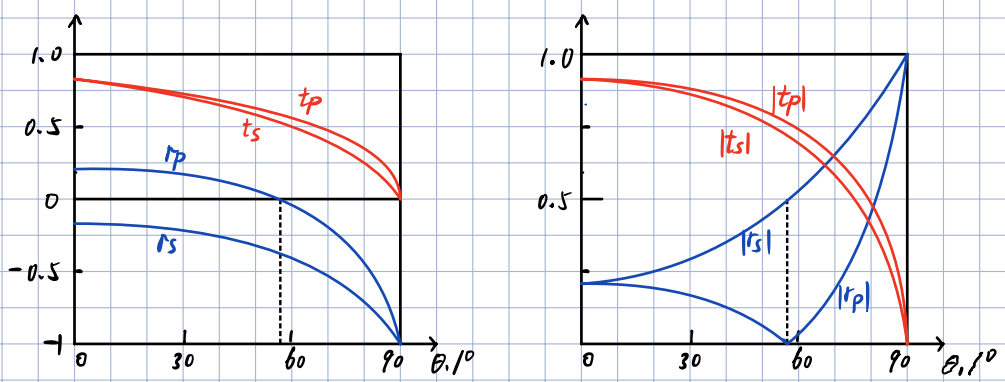
反射光波与透射光波间偏振.

由菲涅尔公式：

$E_{//}$ 与 $E_{\perp}$ 各自独立互不相关.

相应的振幅反射系数 $R_{//}$, $R_{\perp}$ 与透射系数 $T_{//}$, $T_{\perp}$ 不相等

下图为光波在空气与玻璃分界面上的振幅反射与透射系数曲线.



如图，

$|t_p| \geq |t_s|$，　$|r_s| \geq |r_p|$

一般，when 入射为自然光，反射与透射光波均为部分偏振光.

入射为圆偏光，反射与透射光波均为椭圆偏振光.

入射为线偏光，反射与透射均为线偏光.

振动面有一定偏转.

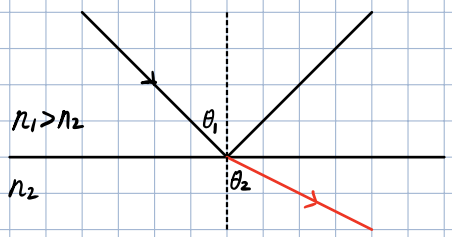
Brewster's law.

由菲涅尔公式，

when $\theta_1 + \theta_2 = 90°$，

$r_p = 0$， $r_s = \cos 2\theta_1$

全反射与倏逝波



$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

when $n_1 > n_2$

$$\sin\theta_2 = \frac{n_1}{n_2}\sin\theta_1$$

when $\sin\theta_1 = \frac{n_2}{n_1}$, 临界角 $\theta_C = \arcsin\frac{n_2}{n_1}$

$\sin\theta_2 = 1$, $\theta_2 = 90°$ 全反射.

全反射能量与能流

由菲涅尔公式：

$$r_s = \frac{E_{1s}'}{E_{1s}} = \frac{\cos\theta_1 - i\sqrt{\sin^2\theta_1 - n_{21}^2}}{\cos\theta_1 + i\sqrt{\sin^2\theta_1 - n_{21}^2}}$$

$$r_p = \frac{E_{1p}'}{E_{1p}} = \frac{n_{21}^2\cos\theta_1 - i\sqrt{\sin^2\theta_1 - n_{21}^2}}{n_{21}^2\cos\theta_1 + i\sqrt{\sin^2\theta_1 - n_{21}^2}}$$

$$|r_s| = \sqrt{r_s \cdot r_s^*} = 1 \qquad |r_p| = 1$$

$$\Rightarrow \qquad |E_{1s}'| = |E_{1s}| \qquad |E_{1p}'| = |E_{1p}|$$

有场，有信息但没有能量.

$$S_{2\delta} = \frac{c}{4\pi}\left(E_{2x}^{\circ} H_{2y}^{\circ} - E_{2y}^{\circ} H_{2x}^{\circ}\right)$$

$$\begin{cases} E_{2x}^{\circ} = \frac{1}{2}\left(E_{2x} + E_{2y}^{*}\right) = -\frac{i}{2}\sqrt{\frac{\sin^2\theta_1}{n_{21}^2} - 1}\left(E_{2p}' e^{-i\tau} - E_{2p}'^{*} e^{i\tau}\right) \\ E_{2y}^{\circ} = \frac{1}{2}\left(E_{1s}' e^{-i\tau} - E_{1s}'^{*} e^{i\tau}\right) \\ \quad = -\frac{i}{2}\sqrt{\varepsilon_2}\sqrt{\frac{\sin^2\theta_1}{n_{21}^2} - 1}\left(E_{2p}' e^{-i\tau} - E_{2p}'^{*} e^{i\tau}\right) \\ H_{2y} = -\frac{1}{2}\sqrt{\varepsilon_2}\left(E_{2s}' e^{-i\tau} + E_{2s}'^{*} e^{i\tau}\right) \end{cases}$$

$\tau = \omega t - k_{2//} x$

$$\frac{1}{2t_0}\int_{-t_0}^{t_0}\left(E_{2p}'^{2} e^{-2i\tau} + E_{2p}'^{*2} e^{i2\tau}\right) dt$$

$-t_0 \le t \le t_0$，$t_0 \gg \tau$.

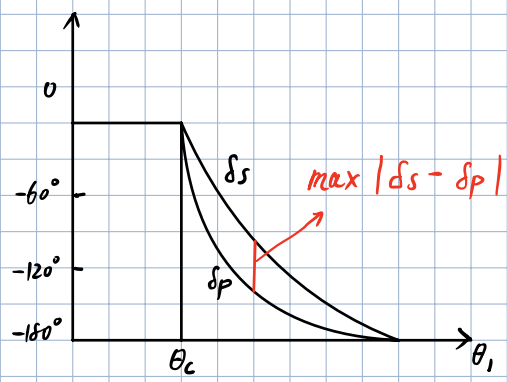
平均能流为0

$$\int_{-t_0}^{t_0}\left(e^{-i2\omega t} + e^{i2\omega t}\right) dt = O\left(\frac{\tau}{t_0}\right) \sim 0$$

相位

$$r_s = e^{i\delta_s} \qquad r_p = e^{i\delta_p}$$

$$\tan\frac{\delta_s}{2} = -\frac{\sqrt{\sin^2\theta_1 - n_{21}^2}}{\cos\theta_1} \qquad \tan\frac{\delta_p}{2} = -\frac{\sqrt{\sin^2\theta_1 - n_{21}^2}}{n_{21}^2\cos\theta_1}$$



$$\tan\frac{\delta_s - \delta_p}{2} = \frac{\cos\theta_1\sqrt{\sin^2\theta_1 - n_{21}^2}}{\sin\theta_1}$$

$$\frac{d}{d\theta_1}\left(\tan\frac{\delta_s-\delta_p}{2}\right) = 0 \Longrightarrow \text{取 } \sin^2\theta_1 = \frac{2n_{21}^2}{1+n_{21}^2}$$

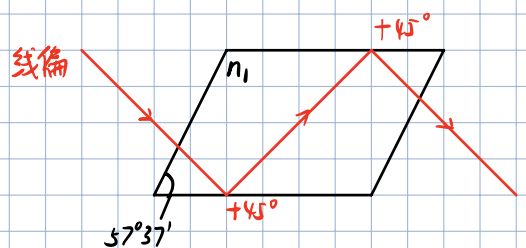
$$\tan\frac{\delta_m}{2} = \frac{1-n_{21}^2}{2n_{21}}$$

线偏 ⟶ 圆偏？

$$\delta_m \geq \frac{\pi}{2} \quad n_{21} \leq \sqrt{2}-1 \leq 0.414 \longrightarrow n_1 \geq 2.41$$

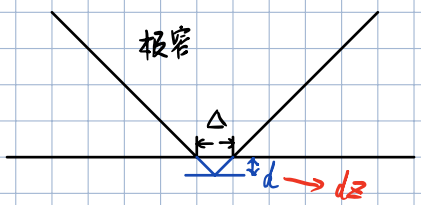
通过2次全反射

$n_1 = 1.51$ $\quad \delta_m = 45°56' > 45°$ $\quad \theta_1 = 48°37'$ $\quad \theta_1' = 54°37'$



菲涅尔菱体.

古斯—汉森位移



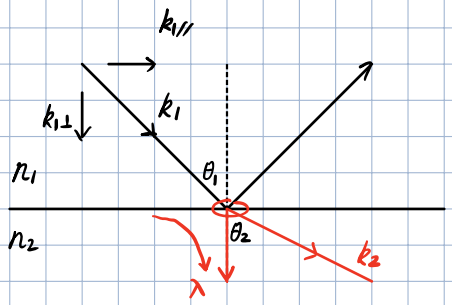
$$\begin{cases} \Delta_S = \dfrac{\lambda_1}{\pi} \cdot \dfrac{\tan\theta_1}{\sqrt{\sin^2\theta_1 - n_{21}^2}} \\ \Delta_P = \dfrac{\lambda_1}{\pi} \cdot \dfrac{\tan\theta_1}{n_{21}\sqrt{\sin^2\theta_1 - n_{21}^2}} \end{cases} \Longrightarrow \quad \Delta_S = 2d_S \tan\theta_1$$

$$d_S = \frac{\lambda_1}{2\pi\sqrt{\sin^2\theta_1 - n_{21}^2}}$$

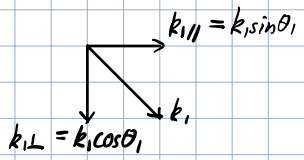
1. $\vec{S}_\perp = 0$

2. $S_{//} \neq 0 \longrightarrow \Delta$

倏逝波

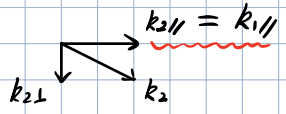


n_1：



$$k_1^2 = k_{1//}^2 + k_{1\perp}^2$$

n_2：



$$k_2^2 = k_{2//}^2 + k_{2\perp}^2$$

$$\frac{k_2}{k_1} = \frac{n_2}{n_1} = n_{21} < 1$$

$$k_{2\perp} = \sqrt{k_2^2 - k_{2//}^2} = k_1\sqrt{n_{21}^2 - \sin^2\theta_1}$$

when $n_{21} < \sin\theta_1$，全反射，$k_{2\perp}$ 变为虚数.

$$k_{2\perp} = ik_1\sqrt{\sin^2\theta_1 - n_{21}^2}$$

$\sqrt{\sin^2\theta_1 - n_{21}^2} = K$

$$\vec{E_2} = \vec{E_{20}} e^{-i(\omega t - \vec{k}\cdot\vec{x})}$$

$$= \vec{E_{20}} e^{-Kz} e^{-i(\omega t - k_{2//} x)}$$

$\vec{E_{20}} = \vec{E_{20}} e^{-Kz}$

振幅非均匀衰减.

定义穿透深度 dz.

$$dz \to \frac{1}{e}|\vec{E_{20}}|$$

$$dz = \frac{\lambda_1}{2\pi\sqrt{\sin^2\theta_{21} - n_{21}}} \quad (\lambda_1 = \frac{\lambda}{n_1})$$

穿透深度约为 1 个波长. 光衰减至没有.

应用: 光子隧道效应.

事实上, 并非仅全反射存在.

倏逝波相速度:

$$v_\varphi = \frac{\omega}{k_{2x}} = \frac{\omega}{k\sin\theta_1} = \frac{v_1}{\sin\theta_1} = \frac{n_2 v_2}{n_1 \sin\theta_1} < v_2.$$

相速度相对一般情形下光在介质中传播更小 —— 慢波.

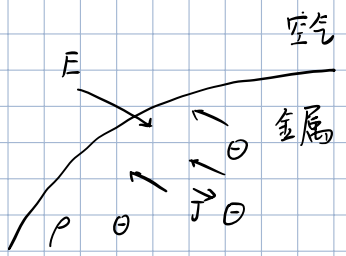
光子隧道效应

金属

1. 导体

σ 电导率 $> 10^4\ (\Omega \cdot m)$

良导体.



① 良导体条件

$$\nabla \cdot \vec{D} = \rho \Rightarrow \nabla \cdot \vec{E} = \frac{\rho}{\varepsilon}$$

ohm's law $\quad \vec{J} = \underline{\sigma}\vec{E}$

$$-\frac{\partial}{\partial t}\rho = \nabla \cdot \vec{J} \longrightarrow \text{电荷守恒}.$$

解得:

$$\rho = \rho_0 e^{-(\frac{\sigma}{\varepsilon}t)}$$

取. $\quad \tau = \frac{\varepsilon}{\sigma} \longrightarrow \rho(\tau) = \frac{1}{e}\rho_0$

光: $T \gg \tau$ $\quad$ 金 10^{-17} $\Rightarrow$ $\omega \ll \frac{\sigma}{\varepsilon}$

$\Rightarrow$ $\quad \frac{\sigma}{\omega\varepsilon} \gg 1$ 良导体条件.

$\Rightarrow$ $\rho_0 = 0$ $\quad \sigma \neq 0$

透射.

Maxwel eqs

$$\begin{cases} \nabla \times E = -\frac{\partial B}{\partial t} \\ \nabla \times H = \sigma E + \frac{\partial D}{\partial t} \\ \nabla \cdot E = 0 \\ \nabla \cdot B = 0 \end{cases}$$

$$\nabla \times (\nabla \times E) = \nabla(\nabla \cdot E) - \nabla^2 E = -\nabla^2 E.$$

$$= -\frac{\partial}{\partial t} \nabla \times B = -\mu\sigma \frac{\partial E}{\partial t} - \varepsilon\mu \frac{\partial^2 E}{\partial t^2}$$

金属良导体波动方程

$$\nabla^2 E - \mu\sigma \frac{\partial}{\partial t} E - \mu\varepsilon \frac{\partial^2}{\partial t^2} E = 0$$

$$E(r,t) = E(r) e^{-i\omega t}$$

$$(\nabla^2 + \omega^2\mu\varepsilon + i\omega\mu\sigma) E(r) = 0.$$

取 $k^2 = \omega^2\mu\varepsilon + i\omega\mu\sigma$

电导率不为0.

k 复数.

$$\nabla^2 E + k^2 E = 0$$

亥姆霍兹方程.

平面波解:

$$\begin{cases} E(r) = E_0 e^{ik\cdot r} \\ B(r) = -\frac{i}{\omega} \nabla \times E(r) \end{cases}$$

取 $k = \beta + i\alpha$

$$E(r,t) = E_0 e^{-\alpha\cdot r} e^{i(\beta\cdot r - \omega t)}$$

振幅指数衰减.

介电常数.

$$\tilde{\varepsilon}_r = 1 + i\frac{\sigma(\omega)}{\omega}$$

$$k^2 = \omega^2\mu\left(\varepsilon + i\frac{\sigma}{\omega}\right) \longrightarrow \tilde{\varepsilon} = \varepsilon + i\frac{\sigma}{\omega} \qquad \tilde{\varepsilon}_r = \varepsilon' + i\frac{\sigma}{\omega\varepsilon_0}$$

Lorentz model. → Drude

$$m\frac{\partial^2}{\partial t^2}\vec{r}(t) + m\Gamma\frac{\partial}{\partial t}\vec{r}(t) + m\omega_0^2\vec{r}(t) = -e\vec{E}(r)$$

$$\vec{r}(t) \longrightarrow \vec{r}(\omega)$$

$$r(\omega) = \frac{e}{m}\frac{E(\omega)}{\omega^2 + i\Gamma^2\omega} \quad \Rightarrow \quad \tilde{\varepsilon}_r = 1 - \frac{Ne^2}{m\varepsilon_0}\frac{1}{\omega^2 + i\Gamma\omega}$$

$$\vec{J}(t) = -Ne\vec{V}(t)$$

$$\vec{v}(t) = \frac{\partial}{\partial t}\vec{r}(t)$$

$$m\frac{\partial}{\partial t}\vec{v}(t) + m\Gamma\vec{v}(t) = -e\vec{E}(t)$$

类似 $\sigma(\omega)$

$$v(t) \overset{FT}{\longleftrightarrow} v(\omega)$$

$$v(t) = \int_{-\infty}^{\infty}\vec{v}(\omega)e^{-i\omega t}d\omega$$

代入:

$$-i\omega m\vec{V}(\omega) + m\Gamma\vec{V}(\omega) = -e\vec{E}(\omega)$$

$$V(\omega) = -\frac{e}{m}\frac{E(\omega)}{\Gamma - i\omega}$$

$$= -i\omega\frac{e}{m}\cdot\frac{\vec{E}(\omega)}{\omega^2 + i\Gamma\omega}$$

$$\vec{J}(\omega) = -Ne\vec{v}(\omega) = \sigma(\omega)\vec{E}(\omega)$$

$$\sigma(\omega) = \frac{Ne^2}{m}\cdot\frac{i\omega}{\omega^2 + i\Gamma\omega}$$

· 考虑 α 与 β.

$$\tilde{k} = \omega^2\mu\tilde{\varepsilon}$$

$$k = \beta + i\alpha$$

$\underline{n}$ 亦为复

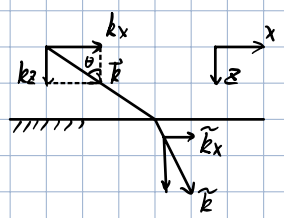
$$\Rightarrow \quad k\cdot k = k^2 = \beta^2 - \alpha^2 + i2\alpha\cdot\beta$$

$$\begin{cases} \beta^2 - \alpha^2 = \omega^2\mu\varepsilon \\ \alpha\cdot\beta = \frac{1}{2}\omega\mu\sigma \end{cases} \qquad \begin{cases} \beta_x^2 + \beta_z^2 - \alpha_z^2 = \omega^2\mu\varepsilon \\ 2\alpha_z\beta_z = \omega\mu\sigma \end{cases}$$

$$\vec{E}(\vec{r},t) = E_0 e^{-i(\omega t - \tilde{k}\cdot\vec{r})} = \boxed{E_0 e^{-\vec{\alpha}\cdot\vec{r}}}\, e^{-i(\omega t - \vec{\beta}\cdot\vec{r})}$$

· 无论如何入射均产生倏逝场.

· 能量进入金属内部. ⟶ 吸收 / 损耗.



$n=1$

金属.

$\lambda \sim 500nm$

$\omega \to 3.8\times10^6\,Hz$.

$\sigma \to 10^7$

$$k_x = \tilde{k}_x = \beta_x + i\alpha_x = k\sin\theta$$

$$\alpha_x = 0$$

垂直入射, $\beta_x = 0$

$$\begin{cases} \beta = \beta_z = \omega\sqrt{\varepsilon\mu}\left\{\frac{1}{2}\left[\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2}+1\right]\right\}^{\frac{1}{2}} \\ \alpha = \alpha_z = \omega\sqrt{\mu\varepsilon}\left\{\frac{1}{2}\left[\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2}-1\right]\right\}^{\frac{1}{2}} \end{cases}$$

$$\alpha \approx \beta = \sqrt{\frac{\omega\mu\sigma}{2}} \qquad \delta = \sqrt{\frac{2}{\omega\mu\sigma}}$$

反射

应用菲涅尔公式. $n \to \tilde{n}$ $\tilde{n} = n + iK$

P光, $$r_p = \frac{n_2\cos\theta_1 - n_1\cos\theta_2}{n_2\cos\theta_1 + n_1\cos\theta_2}$$

S光 $$r_s = \frac{n_1\cos\theta_1 - n_2\cos\theta_2}{n_1\cos\theta_1 + n_2\cos\theta_2}.$$

垂直入射.

$$\left.\begin{aligned} r_p &= \frac{\tilde{n}-1}{\tilde{n}+1} \\ r_s &= -\frac{\tilde{n}-1}{\tilde{n}+1} \end{aligned}\right\} \Rightarrow R = |r_p|^2 = |r_s|^2 = \frac{(n-1)^2+K}{(n+1)^2+k} < 1$$